INCH-POUND

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27 March 1989
SUPERSEDING
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24 January 1974

MILITARY SPECIFICATION

TENT, ASSEMBLY, M-1942

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the erection and sectional components of a circus-type tent fabricated from olive drab, fire, water, weather and mildew-resistant treated cotton cloth. The tent measures approximately 80 feet long by 40 feet wide, with a side wall height of 8 feet and a ridge height of 18 feet.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8340

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SPECIFICATIONS

FEDERAL

T-T-871	- Twine, Cotton, Wrapping
T-T-911	- Twine, Jute
V-T-285	- Thread, Polyester
FF-T-276	- Thimbles, Rope
TT-P-595	- Preservative Coating, Canvas
CCC-C-429	- Cloth, Osnaburg, Cotton
CCC-C-467	- Cloth, Burlap, Jute (or Kenaf)
DDD-L-20	- Label, For Clothing, Equipage, and Tentage, (General Use)
PPP-B-601	- Boxes, Wood, Cleated-Plywood
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner

MILITARY

MIL-C-1410	- Chain, Hook and Ring, Tent
MIL-H-1411	- Hook, Tent, Assembly, Wall Section
MIL-R-1670	- Rope, Tent-Lay
MIL-L-1709	- Lines, Tent
MIL-R-2327	- Rings, Connecting, Round
MIL-C-2399	- Cement, Liquid, Tent Patching
MIL-B-2424	- Block and Tackle, with Load Locking Device
MIL-R-3390	- Rings, Dee
MIL-G-16491	- Grommet, Metallic
MIL-L-40069	- Leather, Cattlehide, Chrome Tanned, Cold-Climate
MIL-T-40625	- Tubing, Bias Sewn (Burlap or Osnaburg), Cloth
MIL-W-43061	- Wax, Machine Stitching
MIL-T-43566	- Tape, Textile, Cotton, General Purpose, Natural or in Colors
MIL-C-43627	- Cloth, Duck, Cotton, Plied Yarns, Fire, Water, Weather and Mildew Resistant Treated, Light Dry Finish

STANDARDS

FEDERAL

FED-STD-751	- Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage

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(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

Department of the Army Field Manual FM-10-16
General Repair of Tents, Canvas and Webbing

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

DRAWINGS

ARMY NATICK LABORATORIES

5-4-99	- Rings, Hand-Worked
5-4-137	- Tent, Assembly, M-1942 Top and Side Views, End Section and Details
5-4-138	- End Section, Enlarged Views A and B and Sections
5-4-139	- Middle Section and Sections
5-4-140	- Enlarged Views C, D, E, F, G, H, and J
5-4-141	- Wall Section, Details and Sections
5-4-142	- Ventilator, Details, Enlarged View K and Sections
5-4-934	- Tent, Assembly, M-1942; Erection Instructions
5-4-948	- Tent Cover (Tent Sections, Frame Type, Maintenance, Medium)
5-4-3359	- Tent Cover

(Copies of drawings are available from U.S. Army Natick Research, Development, and Engineering Center (GL) ATTN: STRNC-EMSS, Natick, MA 01760-5014.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1137.)

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(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Materials and components. Materials and components shall conform to the requirements on the applicable drawings, to subsidiary specifications and standards applicable thereto and as specified herein (see 6.5). It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.2.1 Cloth, duck, cotton. Cloth for the tent shall conform to MIL-C-43627. The color shall be Olive Drab 7.

3.2.2 Tape, textile, cotton. The tape for bindings, tie tapes, and ventilator ring chapes shall conform to type I, class 4, 3/4-inch, Olive Drab 7 of MIL-T-43566.

3.2.3 Thread, polyester. Size 5 thread shall conform to type I or II, class 1, with no finishes applied; size FF and F threads shall conform to type I, class 1, sub-class B, Olive Drab S-1 (CA. 66022) of V-T-285. The dyed thread shall show fastness to weathering equal to or better than the standard shade sample (see 6.4). When no standard shade sample is available, the dyed thread shall show good fastness to weathering. Thread sizes shall be as follows:

For type 301 stitching, except machine stitching reinforcement lines to tent	- Size FF
For machine stitching reinforcement lines to tent	- Size 5
For type 401 stitching indicated by "Z" on drawings	- Size FF for needle, size F for looper

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3.2.4 Rope, manila. The rope used in hems, becket lines, roof reinforcement lines, neck line extension ropes, eave reinforcement lines, neck lines and ridge reinforcement lines shall conform to type I, class 2 of MIL-R-1670 in the sizes indicated on the drawings.

3.2.5 Lines, tent. Tent lines shall conform to class A of MIL-L-1709 as follows:

Tent line type	Use
VI	Wall tie line
IX	Tie line
X	Wall tie line and jumper line
XI	Lacing line, neck, middle section
XX	Lacing line, neck, end section
XXI	Cover tie line
XXVI	Eave line
XXXI	Lacing line, extension cloth
XXXIX	Guy line (on 21-foot pole)

3.2.6 Twine, cotton wrapping. The cotton twine shall conform to type II, class 1, 5-ply of T-T-871.

3.2.7 Leather. Leather for loop protectors shall conform to any type, class 1, 6 ounce, natural color of MIL-L-40069.

3.2.8 Wax, stitching. Wax shall conform to type II, golden color of MIL-W-43061.

3.2.9 Grommets, metallic. Grommets shall conform to type III, class 3, size Nos. 4 and 5 of MIL-G-16491.

3.2.10 Rings, connecting, round. Tent rings shall conform to class 2 or 5, sizes 4 and 6 of MIL-R-2327.

3.2.11 Ring, dee. Dee rings shall conform to configuration B, class 1 of MIL-R-3390.

3.2.12 Thimbles, rope. Thimbles shall conform to type III, sizes 3/8, 1/2, and 3/4 of FF-T-276.

3.2.13 Hook, tent. Tent hooks shall conform to style 1 or 2 of MIL-H-1411.

3.2.14 Chain and ring, tent. Chains and rings shall conform to MIL-C-1410.

3.2.15 Block and tackle. Block and tackles, complete with hoisting lines, shall conform to MIL-B-2424.

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3.3 Construction. The construction shall conform in all respects to the drawings listed in section 2 and as specified herein.

3.3.1 Machine stitching.

3.3.1.1 Types of stitching. All stitch types shall conform to FED-STD-751 as follows:

For all stitching except stitching indicated by "Z" on drawings and stitching reinforcement ropes to tent	- Type 301, 5 to 7 stitches per inch.
For stitching indicated by "Z" on drawings	- Type 401, 5 to 7 stitches per inch. Chain portion of stitching shall not appear on outside of tent and covers, or alternatively - Type 301, 5 to 7 stitches per inch.
For stitching reinforcement ropes to tent	- Type 301, 3 to 5 stitches per inch.

3.3.1.2 Thread breaks. Thread breaks in stitching shall be overstitched not less than 1-inch at each break on stitch type 301 and not less than 1-1/2 inches at each break on stitch type 401. Thread breaks in type 401 stitching may be overstitched with stitch type 301. Any thread breaks noted during inspection shall be repaired by overstitching the existing stitching from a distance of 1 inch before the break to a distance of 1 inch beyond the break. The ends of repair stitching are not required to be backstitched.

3.3.1.3 Stitching ends, type 301. Stitching at the ends shall be overstitched not less than 1-inch except where ends are turned under in a hem, held down by other stitching or where stitching is performed automatically. When stitch patterns such as box, box with cross-stitch, "W" stitching or straight line tacking are performed automatically, at least 3 or more tying, overlapping or backstitches shall be used to secure the ends of stitching.

3.3.1.4 Skipped stitches. Where two or more consecutively skipped stitches occur in type 301 stitching, they shall be overstitched not less than 1-inch. Skipped stitches in type 401 stitching shall be overstitched not less than 1-1/2 inches. Skipped stitches in type 401 stitching may be overstitched with stitch type 301. Skipped stitches detected during inspection shall be repaired as specified for thread breaks (see 3.3.1.2).

3.3.1.5 Lubrication of thread. There shall be no lubrication of the thread by any means prior to or during sewing (see 4.4.2). Except as specified, size 5 polyester thread shall be machine stitched on a machine capable of using the hot stitching wax (see 3.2.8).

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1/2 inch

3.3.2 Handstitching. Handstitching shall be with twine specified in 3.2.6 using four strands twisted together. Each stitch shall be pulled tight and all ends of stitching secured.

3.3.2.1 Rings, hand-worked. Rings shall be handworked as shown on Drawing 5-4-99.

3.3.3 Setting of grommets. Holes punched to receive grommets shall be smaller than the outside diameter of the grommet barrel so that the barrel must be forced through the hole. Grommets shall be securely clinched without cutting the materials.

3.3.4 Splicing of rope. Splicing shall be done as indicated on drawings and as specified in MIL-L-1709.

3.3.4.1 Attaching lines. Eave lines, tie lines, wall tie lines, jumper lines, lacing lines, and extension and neck, center section shall be attached to the tent with a splice as shown on figure 1 or 2 of MIL-L-1709. All such splices shall have a 2-inch eye and three tucks with each strand.

3.3.4.2 Rope in hem. The rope in hem of each wall section may be made of two pieces. When made of two pieces, the ropes shall be joined together with a long splice on opposite ends of the wall section. Such splicings shall be 12 inches long and made as shown on Drawing 5-4-141.

3.3.4.3 Splices, sailmaker's style. The sailmaker's style splices shall be made as shown on figure 10 of MIL-L-1709. The splice shall consist of five tucks, the first three tucks with all three strands of rope, the fourth tuck with two strands, and the fifth tuck with one strand, giving a tapered effect.

3.3.5 Attachment of roof, eave, and ridge reinforcement lines. A fullness of 1/4 inch minimum shall be allowed in the cloth for every foot of rope.

3.3.6 Attachment of hooks, tent, wall section. The hooks shall be assembled in the locations shown on Drawing 5-4-141.

3.3.7 Curtains. The curtains may be fabricated from two or three pieces of duck in length direction. The pieces shall be spliced as shown on Drawing 5-4-141.

3.3.8 Wicking of sewing thread (cup test). There shall be no wicking of water through the tent stitching when the tent is tested as specified in 4.5.1. The cup test is intended to insure that non-wicking thread has been utilized and that no lubrication has been added to the polyester thread during the sewing operation.

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3.3.9 Repairs. All necessary repairs shall be effected in accordance with Department of the Army Field Manual FM-10-16, except that handsewn patches, or cemented patches other than as specified herein, will not be acceptable. Cemented patches on holes, cuts, or tears not more than 1-inch in diameter or length, using cement specified in MIL-C-2399, shall be permitted. A maximum of three cemented patches per tent shall be allowed without prior authorization. All other repairs shall require contracting officer authorization. Repairs required as a result of failure of the cup test shall be performed by treating with a compound conforming to type I, class 1, shade olive drab of TT-P-595, prior to retesting (see 6.9).

3.4 Marking. All marking and labels shall conform to DDD-L-20.

3.4.1 Special marking. When all sectional components of the tent are procured together the marking shall be as follows: The letters "US" and component identification marking on sections and covers shall conform to type IV, class 9 in the size characters and locations shown on drawings unless otherwise specified herein. Class 9 markings shall be clearly legible after subjection to accelerated weathering. Additional identification marking on the appropriate component sections in 1-1/2 inch high letters, centrally located 1 inch below the "US" marking shall be as follows:

a) "TENT, ASSEMBLY, M-1942 WALL SECTION"
b) "TENT, ASSEMBLY, M-1942 MIDDLE SECTION"
c) "TENT, ASSEMBLY, M-1942 END SECTION"

In lieu of the identification marking specified on Drawing 5-4-948, substitute the following:

"TENT, ASSEMBLY, M-1942 1 WALL SECTION"

3.4.1.1 Identification marking. When sectional components, are procured separately they shall be identified as follows:

a) "TENT ASSEMBLY - END SECTION"
b) "TENT ASSEMBY - MIDDLE SECTION"
c) "TENT ASSEMBLY - WALL SECTION"

Use direct marking conforming to the requirements of 3.4.1 with 1-1/2 inch high letters located centrally 1 inch below the "US" marking on the appropriate component.

3.4.2 Identification labels. Identification labels shall conform to type VI, class 6 and shall be 3-1/2 inches by 5-1/2 inches minimum and a blank space of at least 1-1/4 inches shall be provided on the bottom of the label for the Department of Defense inspector's stamp. Identification labels shall be stitched to the end sections, middle sections, wall section and covers in the location shown on the drawings. Fastness shall be as specified for class 6 labels.

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3.4.3 Label of erection instructions. Labels of erection instructions shall conform to type VI, class 7 and shall be stitched to the wall sections and covers in the locations shown on the drawings. The size of the label shall be, borderline to borderline, 12 ± 1/4 inches by 16 ± 1/4 inches with a minimum 1/2-inch margin outside of the borderline on all sides. The information on the label shall be a proportionate copy of Drawing 5-4-934.

3.5 Workmanship. The tent shall conform to the quality of product established by this specification and the occurrence of defects shall not exceed the applicable acceptable quality levels. Cloth components shall be clean and free of holes, cuts, tears, or cloth defects such as multiple floats or broken or missing yarns. Tapes shall contain no frayed or scalloped edges and rope components shall be free of cuts or breaks. Thread tension shall be maintained so that there will be no loose or tight stitching and seam allowances shall be maintained with seams properly sewn so that no runoffs, twists, pleats, or open seams will result. Care shall be taken to insure that needle chews do not occur. All thread ends shall be trimmed to 1/2-inch or less in length. Metal components shall be free of burrs, sharp edges or corroded areas and shall not be broken or malformed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

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4.3 First article inspection. When a first article is required (see 3.1 and 6.3), it shall be examined for the defects specified in 4.4.3 and 4.4.4 and tested for characteristics specified in 4.5.1. The presence of any defect or failure to pass any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.2 In-process inspection. Examination shall be made of the following requirements, operations, or assemblies to establish conformance to specified requirements. Whenever nonconformance is noted, correction shall be made to the items affected, the lot in progress, and to the operation. Parts which cannot be corrected shall be removed from production.

a. There shall be no lubrication of polyester thread, except as specified for size 5 thread, prior to or during sewing operations for compliance with 3.3.1.5.

b. Holes punched to receive grommets shall be in compliance with 3.3.3.

4.4.3 End item visual examination. The end items shall be examined for the defects listed in table I. The lot size shall be expressed in units of completely fabricated tents with covers. The lot size shall be one tent with cover. The tent shall be inspected in an erected position so that the interior is easily accessible for examination. The inspection level shall be I and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be as follows:

For end sections, middle sections, and covers:
10 for major defects
25 for combined major and minor A defects
100 for combined major, minor A, and minor B defects
For wall sections:
6.5 for major defects
15 for combined major and minor A defects
65 for combined major, minor A, and minor B defects

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TABLE I. End item visual defects

Examine	Defect	Classification Major	Minor A	Minor B
Fabric	Broken or missing yarn, multiple floats or open place		201	
	Cut, tear, or hole (including exposed drill hole):			
	-greater than 1/4 inch in length or diameter	101		
	-greater than 1/8 inch, but not greater than 1/4 inch in length or diameter		202	
	-1/8 inch or less in length or diameter:			
	-more than 10 holes		203	
	-more than 5 but not more than 10 holes			301
Tie tapes	Edges frayed or scalloped		204	
Rope components	Any cut, any break	102		
	Kinks, broken or loose ends, bulged strands, strand knots	103		
	Ends not finished as specified		205	
Hardware	Broken or malformed; protective finish omitted; corroded areas; burrs or sharp edges causing damage to item or injuries in handling	104		
Grommets	Improperly set; two or more teeth exposed inside of barrel; loosely clinched, allowing grommet to rotate in hole		206	
	Clinched excessively tight cutting fabric or insecurely clinched; i.e., grommets and washers become disengaged	105		

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TABLE I. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor A	Minor B
Grommets (cont'd)	Size or type not as specified	106		
	Set beyond specified location:			
	-by more than 1/2-inch, but not more than 1 inch			302
	-by more than 1 inch		207	
Thimbles	Misplaced; not spliced into rope as specified	107		
Wall hooks	Misplaced failing to correspond with dee-rings on end and middle section	108		
Seams and stitching				
Open seams	On stitch type 301:			
	-less than 1 inch			303
	-1 inch or more		208	
	On stitch type 401, (except run-offs):			
	-both rows open	109		
	-one row open		209	
	Run-offs on stitch type 401:			
	-more than 1/2 inch, but not more than 1 inch			304
	-more than 1 inch		210	
	NOTE: A seam shall be classified as open when one or more stitches joining a seam are broken, or when one or more skipped or run-off stitches occur. On double-stitched seams, a seam shall be classified as open when either one or both sides of a seam are open.			
	Thread breaks:			
	-overstitched less than 1-1/2 inches on stitch type 401; overstitched less than 1 inch on stitch type 301			305
	NOTE: Thread breaks not overstitched shall be classified as open seams.			

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TABLE I. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor A	Minor B
Seams and stitching (cont'd)				
Open seams (cont'd)	Needle chews resulting in cut, tear, or hole (see cut, tear, or hole)			
	One or more required rows of stitching omitted, except on boxstitching	110		
	Boxstitching incomplete:			
	-one end row of stitching omitted		211	
	-two end rows of stitching omitted	111		
	-any other row of stitching omitted	112		
	Length or width of boxstitching less than specified by more than 1/8 inch			306
	Raw edges caught in stitching (where turnunder is required):			
	-more than 3/4 inch, but not more than 1-1/2 inches			307
	-more than 1-1/2 inches		212	
NOTE:	Raw edges not caught in stitching shall be classified as open seams.			
Seams	Twisted or pleated			308
Seam and stitch type	Wrong seam or stitch type	113		
Stitch tension	Loose, resulting in a loosely exposed top or bottom (bobbin or looper) thread			309
	Tight, resulting in excessive tightness or puckering of fabric		213	
Stitches per inch	One stitch less than minimum specified			310
	Two or more stitches less than minimum specified		214	
	One or two stitches in excess of maximum specified			311
	Three or more stitches in excess of maximum specified		215	

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TABLE I. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor A	Minor B
Seams and stitching (cont'd)				
Stitches per inch (cont'd)	NOTE: Variation in the number of stitches per inch caused by the operator speeding up the machine and pulling the fabric in order to sew over heavy seams or heavy places or turning corners shall be classified as follows: a. Within the minor "A" defect classification -minor "B" defect b. Within the minor "B" defect classification -no defect			
Stitching margin	Larger than specified by more than 3/16 inch: -for more than 2 inches but not more than 4 inches			312
	-for more than 4 inches in length		216	
	Less than specified: -for more than 2 inches but not more than 4 inches in length			313
	-for more than 4 inches in length		217	
Stitching gage	Not as specified		218	
Stitching ends	Overstitched less than 1 inch (except where ends are held down by other stitching or turned under in a hem)		219	
Stitching indicated by "Z" on drawings	Chain portion of stitching shown on outside of tent or cover	114		

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SI and supplement thereto, cited in the specifications and standards (DODISS

TABLE I. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor A	Minor B
Handstitching	Performed with less than four strands of 5-ply twine		220	
Mends, darns, or patches	Repairs not in accordance with Department of the Army Field Manuel FM-10-16, or not authorized by the contracting officer (see 3.3.9)	115		
Hems	Twisted or pleated			314
	Constructed with partial turnunder when double turnunder is required		221	
	Less than specified width: -by more than 1/4 inch, but not more than 1/2 inch			315
	-by more than 1/2 inch		222	
Reinforcement	Improperly applied, causing excessive fullness on reinforcement or reinforced parts			316
	Not securely stitched to reinforced parts on all sides, or not securely caught in seams or stitching of reinforced parts		223	
Binding	Improperly or loosely applied, ends not finished as specified			317
Fullness on cloth	Fullness on cloth omitted along reinforcing rope stitching (see 3.3.5)	116		
Ring chapes	Any part of chapes single-ply construction		224	
Components and assembly	Any component or required operation omitted or improperly applied (unless otherwise classified herein)	117		
Erection instruction labels (wall section and cover)	Omitted, incorrect, illegible, not type specified		225	
Markings; "US" and identification labels	Omitted, incorrect, illegible, or misplaced			318

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TABLE I. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor A	Minor B
Cleanness	Grease or oil stains; thread ends not trimmed to 1/2 inch or less			319
End and middle sections:				
Eave curtain	Lower edge of curtain not scalloped; upper edge not scalloped at eave line opening; upper edge not securely stitched to underside of eave; fullness omitted		226	
	Scalloping uneven or irregular; upper edge uneven with edge of eave by more than 1/8 inch; fullness evenly distributed			320
Eave reinforcement rope	Machine-stitched instead of handstitched to eave underside; handstitching not through curtain, eave facing, and roof; handstitched with less than 1 stitch per turn of strand	118		
Grommet reinforcement	Not securely stitched to body at top and bottom edges; ends not securely caught in eave and neck stitching		227	
Becket edge extension cloth	Misplaced, i.e, (end section) stitched on edge of segment No. 1 (at left of neck) instead of on segment No. 2 (at right of neck); (middle section) stitched on edge at right of neck instead of at left of neck	119		
Beckets	Misplaced, i.e., hand-stitched to edge of segment No. 1 instead of segment No. 2 (end section) stitched at right of neck instead of at left of neck (middle section); top part of beckets not crossed, or not secured by cross stitching	120		

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TABLE I. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor A	Minor B
Beckets (cont'd)	Neck end of becket line not securely spliced into neck line; eave end of becket line not securely spliced into eave reinforcement line		228	
Roof and ridge reinforcement ropes	Not securely stitched to roping seams; long reinforcement ropes on middle section not threaded through ridge reinforcement rope; ends not securely spliced around thimbles	121		
Loops and loop protectors	Opening on roof at location of loop omitted, or failing to provide clearance for pole spindle	122		
	Leather protectors not securely hand-stitched or machine stitched to to loops through eave, eave reinforcement, eave facing, and roof; ends or protectors not secured to loops by hand-stitching or machine stitching		229	
Sidewalls:				
Rope in hem	Omitted, or not continuous all around wall hems	123		
	Not flush with fold line of hem			321
Reinforcement patches	Outer edge of reinforcement patches not caught in both rows of hem stitching		230	
Ventilator flaps	Any ventilator flap misplaced or short, failing to cover opening fully when in a closed position	124		
	Stitched to top of opening instead of bottom		231	

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4.4.4 End item dimensional examination. The end items shall be examined for the defects listed in table II. The lot size shall be expressed in units of tents. The sample unit shall be one completely fabricated tent with cover. The inspection level shall be S-4 and the AQL, expressed in terms of defects per hundred units, shall be 4.0 for major defects, 25 for major and minor A defects combined, and 65 for total (major, minor A, and minor B combined) defects.

TABEL II. End item dimensional defects

Examine	Defect	Classification Major	Minor A	Minor B
Overall dimensions of end, middle and wall sections	Smaller than specified dimensions less applicable minus tolerance indicated on drawings	101		
	Larger than specified dimensions and applicable plus tolerance indicated on drawings			301
Dimensions of cover	Smaller than specified dimensions less applicable minus tolerance indicated on drawings but not smaller than specified dimensions less twice the applicable minus tolerance			302
	Smaller than specified dimensions less twice the applicable minus tolerance indicated on drawings		201	
	Larger than specified dimensions and applicable plus tolerance indicated on drawings			303
Tie tapes	Finished length of tie tapes less than specified by more than 1/2 inch			304
Leather (loop protector)	Less than specified length or width by more than 1/4 inch, thickness less than specified by more than 1/64 inch		202	
Grommets	Eave end grommets on end and middle sections set less than 6-1/2 inches or more than 7-1/2 inches from edge of eave to center of grommet; neck end grommets set less than 1-1/2 inches or more than 2-1/2 inches from edge of neck to			

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TABEL II. End item dimensional defects (cont'd)

Examine	Defect	Classification Major	Minor A	Minor B
Grommets (cont'd)	center of grommet; set in from outer edge of segment less than 1 inch or more than 1-3/4 inches to center of grommets			305
	Intermediate grommets on end, wall and middle sections unequally spaced by more than 3/8 inch	102		
	Wall bottom grommets set up from bottom edge of hem less than 7/8 inch or more than 1-1/4 inches to center of grommet; corner grommets set in from outer edge less than 1 inch or more than 2 inches to center of grommet		203	
Hems	Hems required to finish less than 3/4 inch in width, less than specified; hems required to finish 3/4 inch or more in width, less than specified by more than 1/4 inch		204	
Reinforcements	Finished dimensions less than specified by more than 3/4 inch		205	
Rings and ring chapes	Attached to end and middle section beyond specified location by more than 1/2 inch	103		
Becket edge extension cloth	Less than 8-3/4 inches wide; stitched to edge less than 3-3/4 inches or more than 4-1/2 inches from becket edge		206	
	Eave end of extension cloth set less than 2-1/2 inches or more than 3-1/2 inches from eave edge; neck end set less than 1/2 inch or more than 1 inch from neck edge		207	

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TABEL II. End item dimensional defects (cont'd)

Examine	Defect	Classification Major	Minor A	Minor B
Beckets	Eave end becket formed less than 6-1/2 inches or more than 7-1/2 inches from edge of eave to center of becket; less than 20 inches long; all other beckets except eave end less than 7-3/4 inches long; intermediate beckets unequally spaced by more than 1/2 inch	104		
	Neck end becket formed less than 1-1/2 inches or more than 2-1/2 inches from neck to center of becket		208	
Loops and loop protectors	Loop opening less than 1-1/2 inches or more than 2-1/2 inches long; center of loop less than 2 inches or more than 3 inches from eave edge		209	
Hand-worked rings	Set less than 1-1/4 inches or more than 1-3/4 inches from becket edge to center of ring; set less than 2-1/4 inches or more than 3 inches from eave edge to center of ring		210	
Neck edge	Semi-circular edge irregular in radius; radius less than 6-7/8 inches from point indicated on drawings		211	
Ventilator opening	Cut with vertical center line of opening off center line of panel by more than 1/2 inch; cut with horizontal center line less than 5 feet, 9-1/2 inches or more than 5 feet, 10-1/2 inches from edge of bottom hem; opening less than 8-1/4 inches or more than 9 inches or length, less than 6-1/4 inches or more than 7 inches in width		212	
Ventilator flaps	Stitched to wall less than 3/4 inch or more than 1-1/4 inches from bottom edge of ventilator reinforcement		213	

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4.4.5 End item testing. Tents shall be tested as specified in 4.5.1. The sample unit shall be one completley fabricated tent. The lot size shall be expressed in units of tents. The inspection level shall be S-3. Any test failure shall be cause for rejection of the lot.

4.4.6 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as incomplete closure of cloth wrapping or loose stitching. Bulging or distortion of containers.

4.5 Methods of inspection.

4.5.1 Wicking test. The tent shall be tested in each of three areas as follows: at the peak, ridge, or joining seam abutting the peaks or ridge; at an intersection of a horizontal and vertical seam such as at an intersection of a side wall and roof; and on a roof panel seam. Suspend the seamed section of the test fabric in the center of a 6 ± 1/8 inch diameter hoop and form a depression with the seamed fabric in the hoop to a depth of 1 ± 1/4 inch. Slowly pour 500 mL of water at 77°F ± 4°F into the cupped area (depression) and observe the under surface of the fabric for water penetration. Any wicking of water in any test area along the sewing thread, identified by a discoloration or darkening of the thread within 5 minutes after water is poured, shall constitute a test failure. Water leakage through the needle holes or between the plies of the lapped seamed fabric shall not be considered a failure.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

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5.1.1 Level A.

5.1.1.1 Hardware and tent lines. Each block and tackle, lubricated with the manufacturer's recommended lubricant, shall be completely assembled for use, with the blocks pulled together. Each block and tackle line and each tent guy line shall be neatly coiled and secured by tying the end of each line around itself. The three chain and ring assemblies shall be preserved as specified in MIL-C-1410.

5.1.1.2 Tent sections. The eight tent sections shall be unit packed in six bundles as follows: Two end sections, one per bundle; two middle sections, one per bundle; and four wall sections, two per bundle. The tent sections shall be neatly folded with attached lines placed within the folds. Care shall be exercised during the folding operations to expel all air and to insure that each bundle shall fit its respective cover. Each cover shall be securely closed, taking care that all flaps are neatly tucked in so that no part of the tent is exposed.

5.1.1.3 Component sections procured separately. Each wall section, neatly folded (see 5.1.1.2) and ending with a pack of approximatley 30 inches by 36 inches shall be inclosed in the tent cover detailed on Drawing 5-4-948. End sections and middle sections shall be unit packed as specified in 5.1.1.2.

5.1.2 Commercial. Each tent or component shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing.

5.2.1.1 Hardware and tent lines. The three blocks and tackles, the three chain and ring assemblies, and the nine guy lines, preserved as specified in 5.1, shall be packed together in shipping containers conforming to class 2, style 2 or 4 of PPP-B-621 or overseas type, style A of PPP-B-601. Box and skids (when applicable) shall be fabricated in accordance with load type and gross weight requirements of the box specification. Each shipping container shall be closed and strapped in accordance with the appendix of the applicable container specification.

5.2.1.2 Tent sections. Each bundle, preserved as specified in 5.1, shall be packed in a close-fitting cleated plywood box conforming to overseas type style A of PPP-B-601. Box and skids (when applicable) shall be fabricated in accordance with load type and gross weight requirements of the box specification. Each shipping container shall be closed and reinforced with flat strapping in accordance with the appendix of the box specification.

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5.2.2 Level B packing.

5.2.2.1 Hardware and tent lines. The three blocks and tackles, the three chain and ring assemblies, and the nine tent guy lines, preserved as specified in 5.1, shall be packed together in shipping containers conforming to class 1, style 2 or 4, of PPP-B-621 or domestic type, style A, of PPP-B-601. Box and skids (when applicable) shall be fabricated in accordance with load type and gross weight requirements of the box specification. Each shipping container shall be closed and strapped in accordance with the appendix of the applicable container specification.

5.2.2.2 Tent sections. Each bundle, preserved as specified in 5.1, shall be wrapped and sewn in cotton osnaburg cloth conforming to class 2 of CCC-C-429; or burlap cloth conforming to class 3 of CCC-C-467. The wrapping shall be securely hand-sewn with type I or II, 16-ply cotton twine conforming to T-T-871 or type I, size No. 2 jute twine conforming to T-T-911, with approximately one stitch to the inch and every third stitch overhand knotted. Alternatively, the bottom and side may be machine-sewn. A minimum of 5 inches of wrapping material shall be gathered together on each of the four corners and securely hand-sewn into ears for handles. In lieu of the sewn wrapping indicated above, bias-sewn tubing conforming to MIL-T-40625 may be used. One end of the tubing may be machine-sewn. The formation of handles shall be effected by hand-sewing in conformance with method as described above. Wherever machine sewing is used, the thread type, stitch type and number of stitches per inch shall conform to the requirements specified in MIL-T-40625. However, the thread types and sizes specified for 401 stitching may be used to effect machine sewn closures of bag or tubing material.

5.2.3 Commercial. Tent sections or components, shall be packed in accordance with ASTM D 3951.

5.3 Marking. In addition to any special marking required by the contract or order, shipping containers and cloth covered bundles shall be marked in accordance with MIL-STD-129 or ASTM as applicable.

5.3.1 Special marking. The shipment comprising a complete tent shall be set or assembly marked as specified in MIL-STD-129.

5.3.2 Additional marking. The following shall be stenciled or imprinted on each cloth covered bundle in characters not less than 3/4 inch high:

"SEE ERECTION INSTRUCTIONS

SEWED TO INSIDE OF TENT COVERS"

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6. NOTES

(This sections contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The tent is intended for use in the field for religious services or other purposes such as lectures or the showing of movies. It may also be used for storage, vehicle maintenance shelter, troop quarters or other authorized utility.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required (see 3.1, 4.3, and 6.3).
d. Levels of preservation and packing (see 5.1 and 5.2).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

6.4 Sample. For access to samples, address the contracting activity issuing the invitation for bids.

6.5 Expansion of the tent. By adding middle and wall sections, the length of the tent can be increased by 20 foot increments to any desired length.

6.6 Repair of leaking seams. A satisfactory method of applying the textile compound for the repair of leaking seams is as follows:

a. A suitable volume of textile compound conforming to type I, class 1, shade olive drab of TT-P-595, shall be thoroughly stirred and diluted with an equal volume of petroleum solvent (Stoddard solvent or equivalent). Thoroughly stir the mixture again before use.

b. Prior to application of the compound, the tents must be dry. Special attention must be taken to insure adequate ventilation and that proper fire preventative measures are taken since the compound is flammable.

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c. Two coats of the textile compound should then be applied to the seams (on the outside of the tent only). A 4-inch wide stiff brush is recommended for use. The first coat shall be air dried for a minimum of 4 hours prior to application of the second coat. Elapsed time between coats may be reduced by providing a supply of hot air to evaporate the solvent more quickly. In this manner, a second coat may be applied as soon as the first coat is dry to the touch. After the application of the second coat, the treated tent should be allowed to air dry for a period of at least 24 hours.

6.6.1 Seams to be treated. Seams which have failed the test specified in 4.5.1 should be treated as follows:

a. Failure at roof panel lap seams - treat all roof panel lap seams.

b. Failure at peak reinforcement - treat all peak reinforcement stitching.

c. Failure at eave - treat all stitching at eave and webbing reinforced roof joining seams.

6.7 International standardization agreements. Certain provisions of this document are the subject of international standardization agreement as cited in NATO, STANAG No. 2882, relative to camouflage requirements for tents, shelters and subsidiary components. When amendment, revision, or cancellation of this document is proposed that will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including department standardization offices, if required.

6.8 Subject term (key word) listing.

Circus type
Troop quarters
Vehicle maintenance shelter

6.9 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:

Army - GL
Air Force - 99

Review activities:

Army - MD
Air Force - 82
DLA - CT

Preparing activity:

Army - GL

(Project 8340-0499)

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sed during the folding operations to expel all air and to shall be exposi

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-1110F	Tent, Assembly, M-1942

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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aked in ... on order, shipping containers and cloth covered bundles shall be mar

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